

Name: _____



Exit quiz

Generating a sequence using a position-to-term rule

1 Match up the keywords to their definitions Write the correct letter in each box

a	term
b	term-to-term rule
c	position-to-term rule
d	sequence

	a succession of values or patterns usually following a rule.
	describes how to calculate the term from the term number
	each value or pattern in a sequence
	describes how to get from one term to the next

2 Using the table shown, match the notation to the term. Write the correct letter in each box

Term number	1	2	3	4
Term	3	1	0	-2

a	T_1
b	T_2
c	T_3
d	T_4

	0
	-2
	3
	1



- 3 The position-to-term rule for a sequence is 'multiply the term number by 5 then subtract 10'. What is the 20th term in this sequence? Fill in the blank

- 4 Which of these is a position-to-term rule for the number of dots in each pattern of this sequence? Tick **1** correct answer



- ☐ add 6 to the previous term
- ☐ multiply the term number by 6 then add 4
- ☐ multiply the term number by 6 then add 10
- ☐ Multiply the term number by 10
- ☐ Multiply the term number by 10 then subtract 4

- 5 Which of these is a position-to-term rule for the sequence 12, 22, 32, 42, 52, Tick **1** correct answer

- ☐ the term number add 10
- ☐ multiply the term number by 10
- ☐ multiply the term number by 10 then add 2
- ☐ multiply the term number by 12
- ☐ multiply the term number by 12 and subtract 2



- 6 Work out the 100th term in the constant additive sequence which starts: 13, 21, 29, 37, ...
The first few lines of working are shown. Fill in the blank

$$\begin{array}{ccccccc} 13, & 21, & 29, & 37, & \dots \\ \text{---} & \text{---} & \text{---} & & \\ +8 & +8 & +8 & & \end{array}$$

$$T_1 = 8 + 5$$

$$T_2 = 8 + 8 + 5$$

$$T_3 = 8 + 8 + 8 + 5$$

